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Expandable anterior lumbar implants

Abstract

Expandable fusion devices, systems, and methods thereof. The expandable fusion implant may include upper and lower endplates configured to engage adjacent vertebrae and an actuator assembly for expanding the upper and lower endplates to independently control anterior and posterior heights of the implant. The actuator assembly may be operated in two modes: (1) to force the upper and lower endplates apart resulting in parallel expansion; and (2) to increase the anterior height of the implant resulting in an increase in lordotic angle.

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Primary Examiner: Weiss; Jessica

Background/Summary

FIELD OF THE INVENTION

(1) The present disclosure generally relates to devices and methods for promoting an intervertebral fusion, and more particularly relates to expandable fusion devices capable of being inserted between adjacent vertebrae to facilitate the fusion process and related systems and methods.

BACKGROUND OF THE INVENTION

(2) A common procedure for handling pain associated with intervertebral discs that have become degenerated due to various factors, such as trauma or aging, is the use of intervertebral fusion devices for fusing one or more adjacent vertebral bodies. Generally, to fuse the adjacent vertebral bodies, the intervertebral disc is first partially or fully removed. An intervertebral fusion device is then inserted between neighboring vertebrae to maintain normal disc spacing and restore spinal stability, thereby facilitating an intervertebral fusion.

(3) There are a number of fusion devices and methodologies for accomplishing the intervertebral fusion. These may include solid bone implants, fusion devices which include a cage or other implant mechanism, which may be packed with bone and/or bone growth inducing substances, and expandable implants. The expandable implants may be inserted into the intervertebral disc space at a minimized height, and then expanded to restore height loss in the disc space.

(4) There are drawbacks, however, with existing expandable implants including excessive impaction during insertion, visual obstruction, and imperfect matching with patient's lordosis due to discrete increments in lordotic angulation. As such, there exists a need for fusion devices capable of providing distraction as well as achieving optimal height restoration and changes in lordotic angulation independently from its expansion.

SUMMARY OF THE INVENTION

(5) To meet this and other needs, implants, systems, and methods for performing intervertebral fusion and spine stabilization are provided. In particular, expandable intervertebral implants, for example, for anterior spinal surgery may be used to treat a variety of patient indications. The expandable implants are configured to be inserted into the intervertebral disc space at a minimized height, and then expanded axially to restore normal spinal alignment and distribute the load across

the vertebral endplates. The implant may provide distraction as well as achieving optimal height restoration. The implant may also change in lordotic angulation independently from its expansion.

(6) According to one embodiment, an expandable implant includes an upper endplate and a lower endplate configured to engage adjacent vertebrae and an actuator assembly for expanding the upper and lower endplates to independently control anterior and posterior heights of the implant. The actuator assembly includes a moveable anterior actuator, a moveable posterior actuator, and a stationary posterior actuator positioned between the upper and lower endplates. An actuator screw threads into an anterior actuator nut located in the moveable anterior actuator and a posterior actuator nut located in the stationary posterior actuator. The actuator screw threads through the moveable posterior actuator to translate the posterior actuator. When the actuator screw and anterior actuator nut are turned together, the moveable posterior actuator moves toward the stationary posterior actuator, forcing the upper and lower endplates apart resulting in parallel expansion. When only the anterior actuator nut is turned, the moveable anterior actuator moves alone to increase the anterior height, resulting in an increase in lordotic angle.

(7) The expandable implant may include one or more of the following features. The upper and lower endplates may define a plurality of ramps configured to mate with corresponding ramps on the anterior and posterior actuators. The ramps on the upper and lower endplates may include female T-slots. The ramps on the anterior and posterior actuators may include male protrusions with point contact pivoting ramps, facilitating pivoting or rotational movement. The moveable anterior actuator may include a laterally extending body with an enlarged central portion defining a non-threaded bore and free ends defining an irregular cross-sectional shape with a pair of male ramps extending from each end of the anterior actuator. The moveable posterior actuator may include a laterally extending body defining a threaded bore with a flat anterior face and an opposite posterior face defining point contact pivoting ramps configured to interface with the corresponding ramps of the upper and lower endplates. The stationary posterior actuator may include a laterally extending body defining a non-threaded bore with an anterior face defining point contact pivoting ramps configured to interface with the corresponding ramps of the upper and lower endplates. The actuator screw may include a shaft having a single type of thread profile along its length and a reduced diameter at its distal end to thread into the posterior actuator nut. The single type of thread profile achieves both parallel and lordotic expansion.

(8) According to another embodiment, an expandable implant includes an upper endplate and a lower endplate configured to engage adjacent vertebrae and an actuator assembly for expanding the upper and lower endplates to independently control anterior and posterior heights of the implant. The actuator assembly includes a moveable anterior actuator securing a plurality of anterior pivot ramps, a moveable posterior actuator and a stationary posterior actuator engaged with upper and lower posterior pivot ramps, and an actuator screw threaded into an anterior actuator nut located in the moveable anterior actuator and a posterior actuator nut located in the stationary posterior actuator. The actuator screw threads through the moveable posterior actuator to translate the posterior actuator. When the actuator screw and the anterior actuator nut are turned together, the moveable posterior actuator moves toward the stationary posterior actuator, forcing the posterior pivot ramps outward resulting in parallel expansion of the upper and lower endplates. When only the anterior actuator nut is turned, the moveable anterior actuator moves alone to increase the anterior height, resulting in an increase in lordotic angle.

(9) The expandable implant may include one or more of the following features. The upper and lower endplates may define a plurality of ramps configured to mate with corresponding ramps on the anterior and posterior pivot ramps. Each anterior pivot ramp may have a ring received on the moveable anterior actuator and a foot with a sliding surface configured to mate with respective ramps on the upper and lower endplates. The foot may be a male projection extending from one side of the ring, and the ramp on the upper and lower endplates may be a female recess configured to receive the male projection of the foot. The moveable anterior actuator may include a laterally

extending body with an enlarged central portion defining a non-threaded bore and free ends defining cylindrical ends. Each anterior pivot ramp may have a smooth inner surface to allow each anterior pivot ramp to rotate on the cylindrical ends of the moveable anterior actuator. The posterior pivot ramps may be separated into left and right sections positioned on opposite sides of the actuator screw. The posterior pivot ramps may include a lateral extending body with a bell-shaped cross section having a tapered end and flared end. The flared end of the posterior pivot ramps may include a lateral rib having a circular cross section configured to pivot in a corresponding channel in the upper and lower endplates.

(10) According to another embodiment, a method of spinal fixation includes: inserting an expandable intervertebral implant into a disc space between adjacent vertebrae, the expandable implant comprising upper and lower endplates and an actuator assembly including a moveable anterior actuator, a moveable posterior actuator, and a stationary posterior actuator positioned between the upper and lower endplates, an actuator screw threads into an anterior actuator nut located in the moveable anterior actuator and a posterior actuator nut located in the stationary posterior actuator, the actuator screw threads through the moveable posterior actuator to translate the posterior actuator; and expanding the implant in height by (1) rotating the actuator screw and anterior actuator nut together to move the moveable posterior actuator toward the stationary posterior actuator, thereby forcing the upper and lower endplates apart resulting in parallel expansion; or (2) rotating only the anterior actuator nut to move the moveable anterior actuator alone to increase the anterior height, resulting in an increase in lordotic angle. The method may also include inserting bone anchors through sockets in the upper and lower endplates and into the adjacent vertebrae, and rotating blocking screws in the upper and lower endplates to cover the bone anchors and prevent the bone anchors backing out.

(11) According to yet another embodiment, a kit may include a plurality of expandable implants of different sizes and configurations, fasteners or anchors, k-wires, and other components for performing the procedure. The kit may further include one or more devices suitable for installing and/or removing the implants, such as insertion devices or drivers, expansion instruments, removal devices, and other tools and devices which may be suitable for surgery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present embodiments will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIGS. **1**A-**1**B illustrate perspective and side views, respectively, of an expandable implant in a collapsed position according to one embodiment;

(3) FIGS. **2**A-**2**B illustrate perspective and side views, respectively, of the expandable implant in a fully expanded position according to one embodiment;

(4) FIGS. **3**A-**3**B illustrate perspective and side views, respectively, of the expandable implant in an expanded position with the anterior and posterior height of the implant independently controlled according to one embodiment;

(5) FIG. **4** illustrates an exploded view of the expandable implant according to one embodiment;

(6) FIGS. **5**A-**5**B illustrate anterior or rear views of the expandable implant in the collapsed and fully expanded positions, respectively, according to one embodiment;

(7) FIGS. **6**A-**6**B illustrate vertical cross-sectional views of the expandable implant in the fully expanded position according to one embodiment;

(8) FIG. **7** shows an axial cross-sectional view of the expandable implant according to one embodiment;

(9) FIGS. **8**A-**8**B illustrate perspective and side views, respectively, of an expandable implant in a

collapsed position according to one embodiment;
(10) FIGS. **9**A-**9**B illustrate perspective and side views, respectively, of the expandable implant in a fully expanded position according to one embodiment;
(11) FIGS. **10**A-**10**B illustrate perspective and side views, respectively, of the expandable implant in an expanded position with the anterior and posterior height of the implant independently controlled according to one embodiment;
(12) FIG. **11** illustrates an exploded view of the expandable implant according to one embodiment;
(13) FIGS. **12**A-**12**B illustrate anterior or rear views of the expandable implant in the collapsed and fully expanded positions, respectively, according to one embodiment; and
(14) FIGS. **13**A-**13**B show cross-sectional side and top views, respectively, of the expandable implant according to one embodiment.

DETAILED DESCRIPTION

(15) Embodiments of the disclosure are generally directed to devices, systems, and methods for intervertebral fusion and spine stabilization. Specifically, expandable implants are configured to be inserted into the intervertebral disc space at a minimized height, and then expanded axially to restore normal spinal alignment and distribute the load across the vertebral endplates. The implant may provide distraction as well as achieving optimal height restoration. The implant may also change in lordotic angulation independently from its expansion.
(16) A spinal fusion is typically employed to eliminate pain caused by motion of degenerated disc material. Upon successful fusion, a fusion device becomes permanently fixed within the intervertebral disc space. The expandable fusion device may be positioned between adjacent vertebral bodies in a collapsed position. The expandable fusion device is configured to expand in height to restore height loss in the disc space. The fusion device engages the endplates of the adjacent vertebral bodies and, in the installed position, maintains desired intervertebral disc spacing and restores spinal stability, thereby facilitating the intervertebral fusion.
(17) Minimally invasive surgery (MIS) may be used to preserve muscular anatomy by only causing disruption where necessary. The benefit of the MIS surgical approach is that it can reduce post-operative pain and improve recovery time for patients. In one embodiment, the expandable fusion device can be configured to be placed down an endoscopic tube and into the surgical target site. By way of example, the surgical site may be an intervertebral disc space situated between two adjacent vertebrae. Although particularly suited for use in an anterior lumbar interbody fusion (ALIF), it will be readily appreciated by those skilled in the art that the implant may be employed in any number of suitable orthopedic approaches and procedures, such as direct lateral where coronal deformity is encountered. Other approaches may include but are not limited to posterior, lateral, anterolateral, posterolateral, or transforaminal approaches to the lumbar spine, cervical spine, or thoracic spine, as well as any non-spine application, such as treatment of bone fractures and the like. The terms implant, interbody, interbody implant, fusion device, spacer, and expandable device may be used interchangeably herein.
(18) Components of all of the devices disclosed herein may be manufactured of any suitable materials including metals (e.g., titanium), metal alloys (e.g., stainless steel, cobalt-chromium, and titanium alloys), ceramics, plastics, plastic composites, or polymeric materials (e.g., polyether ether ketone (PEEK), polyphenylene sulfone (PPSU), polysulfone (PSU), polycarbonate (PC), polyetherimide (PEI), polypropylene (PP), polyacetals, or mixtures or co-polymers thereof), and/or combinations thereof. In some embodiments, the devices may include radiolucent and/or radiopaque materials. The components can also be machined and/or manufactured using any suitable techniques (e.g., 3D printing).
(19) Turning now to the drawing, where like reference numerals refer to like elements, FIG. **1**A-**7** illustrate an expandable fusion device or implant **10** according to one embodiment. The expandable implant **10** extends along a central longitudinal axis A between a front end or posterior end **12** and a rear end or anterior end **14** of the device **10**. The implant **10** includes upper and lower endplates

16, **18** configured to engage adjacent vertebrae, which define a height of the implant **10**. The implant **10** includes an actuator assembly **20** configured for expanding the upper and lower endplates **16**, **18**. The actuator assembly **20** includes a single actuator screw **22** threaded into anterior and posterior actuator ramps **28**, **30** to independently control the anterior and posterior height of the implant **10**.

(20) The implant **10** may include three separate actuators **28**, **30**, **32** positioned between the upper and lower endplates **16**, **18**: a moveable anterior actuator **28**, a moveable posterior actuator **30**, and a stationary posterior actuator **32**. One end of the actuator screw **22** threads into an anterior actuator nut **24** located in the anterior actuator **28** and the opposite end of the actuator screw **22** threads into a posterior actuator nut **26** affixed to the stationary posterior actuator **32**. The actuator screw **22** threads through the moveable posterior actuator **30** to translate the posterior actuator **30** along longitudinal axis A. When the actuator screw **22** and anterior actuator nut **24** are turned together, the moveable posterior actuator **30** moves toward the stationary posterior actuator **32**, forcing the upper and lower endplates **16**, **18** apart resulting in parallel expansion. When only the anterior actuator nut **24** is turned, the moveable anterior actuator **28** moves alone to increase the anterior height, resulting in an increase in lordotic angle.

(21) The implant **10** is configured to be inserted into the disc space in a collapsed configuration. FIGS. **1**A-**1**B show the implant **10** in the collapsed configuration. Once inserted into the disc space, the implant **10** is expanded in height to an expanded configuration to precisely restore spinal alignment and distribute load across the vertebral endplates. FIGS. **2**A-**2**B show the implant **10** in a fully expanded configuration. FIGS. **3**A-**3**B show the implant **10** in an expanded configuration where the anterior and posterior heights are independently adjustable to a desired lordotic profile. In this manner, the height is adjustable to restore height loss in the disc space and lordotic angulation. It should be understood that reference to the front and rear ends and anterior and posterior heights are described with respect to the direction of placement into an intervertebral disc space with the front of the expandable fusion device **10** placed into the disc space first, followed by the rear of the expandable fusion device **10**, and then expanding the endplates **16**, **18** in height and/or lordosis. These and other directional terms may be used herein for descriptive purposes and do not limit the orientation(s) in which the devices may be used.

(22) Each endplate **16**, **18** may include a front or posterior rail **40** and a rear or anterior rail **42** extending between opposed side rails **44**, **46**. The rails **40**-**46** define an inner face **50** and an opposite outer face **52**. The inner face **50** may be configured to mate with the respective actuators **28**, **30**, **32** and the outer face **52** may be configured to contact adjacent vertebrae. The outer face **52** of each endplate **16**, **18** may include a plurality of teeth **54** or other friction increasing elements, such as ridges, roughened surfaces, keels, gripping or purchasing projections configured to retain the device **10** in the disc space. The endplates **16**, **18** may be 3D printed using additive manufacturing. In this manner, the outer face **52** may be created with teeth and/or surface texturing that can better facilitate bony on-growth. Each endplate **16**, **18** may define a vertical window or through passage **56**, thereby defining a central graft chamber within the implant **10**. The window or through passage **56** allows graft material or other therapeutically beneficial material to be packed into or grow through the implant **10**.

(23) The implant **10** may be secured to the adjacent vertebrae with one or more anchors or fixation screws (not shown). For example, the anterior rail **42** may define at least one anchor socket **60** configured for receiving the anchor or fixation screw therethrough and into the adjacent vertebra. In the embodiment shown, three sockets **60** for receiving three respective anchors or screws are provided in the upper and lower endplates **16**, **18**: one socket **60** pointed upward into the superior vertebra and two sockets **60** pointed downward into the inferior vertebra. The sockets **60** may be surrounded by a hemispherical protrusion such that the anchors or screws may be angled into the adjacent vertebrae. In one embodiment, the bone screws may be polyaxial screws, and sockets **60** correspondingly shaped, such that the polyaxial screws may be inserted at optimal angles with

respect to implant **10**. In another embodiment, the anchor may be a curved, t-shaped shim-type anchor with sharp edges to penetrate bone. Examples of bone screws and anchors are further described in in U.S. Pat. No. 11,554,023, which is incorporated by reference herein in its entirety for all purposes. Although a given configuration of sockets **60** is shown, it will be appreciated that the sockets **60** may be present in any suitable number and configuration for fixation. In the alternative, the sockets **60** may be omitted to provide a standalone device.

(24) Cam style blocking screws **62** may be used to block the anchors or fixation screws from backing out after being inserted. The anterior rail **42** may define a blocking screw hole **64** positioned next to each respective socket **60**. The blocking screw holes **64** may be internally threaded to receive the respective threaded blocking screws **62**. In one embodiment, three blocking screws **62** thread into the endplates **16**, **18** to secure the anchors or fixation screws in the three respective sockets **60**. The blocking screws **62** may have an enlarged head with a drive recess and a threaded shaft. When the blocking screws **62** are rotated and engaged, a portion of the enlarged head covers the respective anchors or fixation screws, thereby preventing migration of the installed anchors or fixation screws.

(25) As best seen in FIG. **5**A, the implant **10** may include one or more recesses or

(26) openings **66** for receiving an instrument, such as an insertion and/or expansion instrument. The anterior rail **42** of the lower endplate **18** may define a first opening **66** on one lateral side and a second opening **66** on the opposite side of the lower endplate **18** configured to be retained by an insertion instrument. For example, the first opening **66** may be a threaded cylindrical opening configured to receive a threaded portion of the inserter instrument and the second opening **66** may be a non-threaded non-cylindrical recess configured to receive a non-threaded portion of the inserter instrument. It will be appreciated that the implant **10** may connect with an insertion and/or expansion instrument in any suitable manner.

(27) The inner face **50** of each endplate **16**, **18** may include one or more ramps **70**, **72** configured to mate with the respective actuators **28**, **30**, **32**, thereby causing endplates **16**, **18** to separate apart. For example, the endplates **16**, **18** may define posterior ramps **70** along the inner face **50** of the posterior rail **50** and anterior ramps **72** along the inner face **50** of the side rails **44**, **46**. The ramps **70**, **72** may include ramped surfaces, angled surfaces, or inclined planes with a given gradient or angle of slope. The ramps **70**, **72** may have generally straight ramped surfaces, may be curved, or may be configured in any suitable manner for slidable interface between the components. The ramps **70**, **72** may define male or female slide ramps configured to mate with corresponding ramps **90**, **112**, **132** on the actuators **28**, **30**, **32**. In one embodiment, the ramps **70**, **72** on endplates **16**, **18** include female ramps with grooves or slots **74** defined therein, such as T-slots. The groove or slots **74** may be configured to receive a portion of point contact pivoting ramps **90**, **112**, **132** of the actuators **28**, **30**, **32**. As the anterior and/or posterior actuators **28**, **30**, **32** translate and slide against the ramps **70**, **72** of the endplates **16**, **18**, the movement provides for expansion or contraction of the implant **10**. The expansion may include the ability to individually adjust the anterior and/or posterior heights of the implant **10**.

(28) The actuator assembly **20** includes moveable anterior actuator **28** positioned between the upper and lower endplates **16**, **18**, thereby providing anterior expansion to the implant **10**. The moveable anterior actuator **28** includes a laterally extending body with an enlarged central portion **80**. The anterior actuator **28** extends from a first free end **82** to a second free end **84**. The first and second ends **82**, **84** may define an irregular cross-sectional shape, such as a polygon with facets and rounded corners. The first and second free ends **82**, **84** are receivable between the side rails **44**, **46** toward the anterior end **14** of the upper and lower endplate **16**, **18** and the enlarged central portion **80** is positionable through the graft window **56** of the upper and lower endplates **16**, **18** when in the collapsed configuration. The enlarged central portion **80** defines a central non-threaded bore **86** sized and dimensioned to receive the anterior actuator nut **24**. The central axis of bore **86** may be aligned with the central longitudinal axis A of the implant **10**. Additional recesses or bores **88** may

be provided through the anterior actuator **28**. For example, a first threaded bore **88** may be provided on one side of the central bore **86** and a second non-threaded bore **88** may be provided on the opposite side of the central bore **86**. Recesses or bores **88** may be used to attach an instrument, such as an insertion and/or expansion instrument.

(29) The moveable anterior actuator **28** includes one or more ramps **90** configured to mate with corresponding anterior ramps **72** on the endplates **16**, **18**. The ramps **90** may include two point contact pivoting ramp surfaces incorporated into each side of the anterior actuator **28**. For example, the anterior actuator **28** may define a pair of male ramps **90** proximate each end **82**, **84** of the actuator **28** and configured to mate with the corresponding anterior ramps **72** along the side rails **44**, **46** of the endplates **16**, **18**. The point contact pivoting ramp **90** may be configured to interface corresponding anterior ramp **72** at a singular contact point or location, facilitating pivoting or rotational movement. The singular location where the ramp **90** interacts with corresponding ramp **72** may be a focal point around which rotation or movement occurs. Unlike broad surface contact between mating ramps, ramp **90** may have targeted contact at a defined point or edge that interfaces with corresponding ramp **72**. The engaging surface of the male ramp **90** may be pointed, angled, tapered, curved, etc. to define a single point or edge of contact.

(30) In one embodiment, the male ramps **90** may include projections or protrusions with a rail configured to engage with the corresponding grooves or slots **74** in the endplates **16**, **18**. In some embodiments, the protrusions or rails may include L-shaped tabs which extends out from the actuator **28** and then make a 90-degree turn forming the L shape. In other embodiments, the protrusion or rails may include T-shaped tabs with a stem which extends out from the actuator **28** and then provides a perpendicular extension at the free end of the stem forming the top of the T shape. The free ends of the tabs may point laterally outward and away from one another or in any suitable manner. The slots **74** machined into the upper and lower endplates **16**, **18** interface with the point contact pivoting ramps **90** on the anterior actuator **28** allowing for parallel and lordotic expansion, while preventing disassembly by pulling the endplates apart. Two limit pins **92** may thread into the anterior actuator **28** to prevent the implant **10** from being disassembled if over expanded. The point contact pivoting ramps **90** on the anterior actuator **24** translate axial motion from the actuator screw **22** into anterior expansion, and the ability of the endplate slots **74** to pivot about the point contact allows for independent expansion of the anterior and posterior actuators **28**, **30**.

(31) The actuator assembly **20** includes a moveable posterior actuator **30** positioned between the upper and lower endplates **16**, **18**, thereby providing posterior expansion to the implant **10**. The moveable posterior actuator **30** includes a laterally extending body connecting a first free end **102** to a second free end **104** with a flat anterior face **106** and an opposite posterior face **108**. The first and second free ends **102**, **104** are receivable between the upper and lower endplates **16**, **18** toward the posterior end **12** of the implant **10**. The posterior actuator **30** defines a central threaded cylindrical bore **110** configured to receive the actuator screw **22**. The central axis of bore **110** may be aligned with the central longitudinal axis A of the implant **10**. As best seen in FIG. **1**B, in the collapsed position, the flat anterior face **106** may be configured to contact or rest against the posterior face of the central portion **80** of the anterior actuator **28**.

(32) The moveable posterior actuator **30** includes one or more ramps **112** configured to mate with corresponding ramps **70** on the endplates **16**, **18**. The ramps **112** may be defined into and/or extend from the posterior face **108** of the actuator **30**. Similar to ramps **90**, the ramps **112** may include a point contact pivoting ramp surface incorporated into each side of the moveable posterior actuator **30**. For example, the moveable posterior actuator **30** may define a pair of male ramps **112** at each end **102**, **104** of the actuator **28** and configured to mate with the posterior ramps **70** along the posterior rail **40** of the endplates **16**, **18**. Two male ramps **112** may be positioned upward to interface with the upper endplate **16** and two male ramps **112** may be positioned downward to interface with the lower endplate **18**. The male ramps **112** may include projections or protrusions

having a L-shaped tabs configured to engage with the corresponding grooves or slots **74** in the endplates **16**, **18**. The free ends of the L-shaped or T-shaped tabs may point laterally and away from one another on opposite sides **102**, **104** of the actuator **30**. The point contact pivoting ramps **112** on the moveable posterior actuator **30** interface with the corresponding slots **74** in the upper and lower endplates **16**, **18**, preventing disassembly and allowing for independent expansion.

(33) The actuator assembly **20** includes a stationary posterior actuator **32** positioned between the upper and lower endplates **16**, **18**, thereby providing posterior expansion to the implant **10**. The stationary posterior actuator **32** includes a laterally extending body having a first free end **120** and a second free end **122** on opposite sides. The first and second free ends **120**, **122** are receivable between the posterior rails **40** of the upper and lower endplate **16**, **18**, and the stationary posterior actuator **32** defines the nose or front end **12** of the implant **10**. The stationary posterior actuator **32** includes an anterior face **124** and an opposite posterior face **126**. A central block **128** may protrude from the anterior face **124** which defines a central non-threaded bore **130** sized and dimensioned to receive the posterior actuator nut **26**. The central axis of bore **130** may be aligned with the central longitudinal axis A of the implant **10**.

(34) The posterior expansion mechanism functions by using the actuator screw **22** to drive the moveable posterior actuator **30** toward the stationary posterior actuator **32**. As best seen in FIG. **2**B, the moveable posterior actuator **30** translates away from anterior actuator **28** and posteriorly toward stationary posterior actuator **32**. As the two posterior actuators **30**, **32** move toward one another, these posterior actuators **30**, **32** interface with ramped surfaces **70** and slots **74** on the upper and lower endplates **16**, **18**, pushing the upper and lower endplates **16**, **18** out in both directions, expanding the posterior height of the implant **10**.

(35) The stationary posterior actuator **32** includes one or more ramps **132** configured to mate with corresponding posterior ramps **70** on the endplates **16**, **18**. The ramps **132** may be defined into and/or extend from the anterior face **124** of the actuator **32**. Similar to ramps **90**, **112**, the ramps **132** may include point contact pivoting ramp surfaces. For example, the posterior actuator **32** may define a male ramp **132** near each end **120**, **122** of the actuator **32** configured to mate with the posterior ramps **70** along the posterior rails **40** of the endplates **16**, **18**. A pair of posterior ramps **132** may be positioned near each end **120**, **122** with two configured to engage the upper endplate **16** and two configured to engage the lower endplate **18**. The male ramps **132** may include projections or protrusions having a L-shaped or T-shaped tabs configured to engage with the corresponding grooves or slots **74** in the endplates **16**, **18**. The free ends of the tabs may point laterally and away from one another. When the moveable posterior actuator **30** moves toward the stationary posterior actuator **32**, these posterior actuators **30**, **32** interface with ramped surfaces **70** and slots **74** on the upper and lower endplates **16**, **18**, pushing the upper and lower endplate **16**, **18** out in both directions, thereby expanding the posterior height of the implant **10**.

(36) Instead of accommodating lordotic angle adjustment by allowing the engagement features (e.g., T-slots and rails) to rotate about a set axial pivot point in the anterior and posterior actuators **28**, **30**, **32**, implant **10** uses a point contact on the engagement interface itself as a pivot point for lordotic angle adjustment. The point contact engagement interface **90**, **112**, **132** includes enough angular clearance with the mating engagement slot **74** to accommodate the endplate angle change required for in-situ lordotic adjustment and device assembly.

(37) The actuator assembly **20** includes actuator screw **22**, anterior actuator nut **24**, and posterior actuator nut **26** aligned along the central longitudinal axis A of the implant **10**. The actuator screw **22** extends from a proximal end **140** to a distal end **142**. The actuator screw **22** may include a shaft with external threads(s) or an exterior threaded portion **144** extending along its length. The threaded shaft **144** may have a given diameter, handedness, thread form, thread angle, lead, pitch, etc. suitable for interfacing with both the anterior actuator nut **24** and the moveable posterior actuator **30**. In this embodiment, a single type of thread profile is used for both parallel and lordotic expansion. Other systems may require three different sets of threads to engage for parallel

expansion. Conversely, threaded shaft **144** achieves parallel and lordotic expansion through the interaction of one set of threads. This reduces the overall increase in lifting torque created by friction or binding of the threads.

(38) The proximal end **140** of shaft **22** may define an instrument recess **146** configured to receive an instrument, such as a driver, to rotate or actuate the actuator screw **22**. The instrument recess **146** may include a tri-lobe, hex, star, or other suitable recess configured to engage with a driver instrument to apply torque to the actuator screw **22**. The proximal end **140** of the shaft is configured to thread into the anterior actuator nut **24** to translate the moveable anterior actuator **28**. The threaded shaft **144** also threads through the moveable posterior actuator **30** to translate the moveable posterior actuator **30** along central longitudinal axis A.

(39) The distal end **142** of the actuator screw **22** may have a reduced distal tip, for example, having a diameter less than the diameter of the threaded shaft **144**. The distal tip **142** may be threaded and configured to thread into the posterior actuator nut **26** retained in the stationary posterior actuator **32**. Even at full expansion, for example shown in FIGS. **6**A-**6**B, the distal end **142** of the actuator screw **22** does not protrude out past the posterior edge or front nose **12** of the implant **10** as may be seen in other designs. Posterior protrusion of the actuator screw was a point of concern from surgeons due to the optics of having the rod protruding near the posterior structures of the spine. Thus, the expansion assembly **20** and operation for implant **10** eliminates the possibility of any posterior protrusion of the actuator screw **22**.

(40) The anterior actuator nut **24** has a body extending between a proximal end **150** to a distal end **152**. As best seen in FIG. **6**A, a central bore **154** extends through the body of the anterior actuator nut **24** from the proximal end **150** to the distal end **152**. A portion of the bore **154** defines internal threads which are configured to threadably engage with the exterior threads **144** of the actuator screw **22**. The proximal end **150** of the anterior actuator nut **24** defines a driver engagement recess, such as a series of notches and teeth or other suitable recess configured to engage with a driver instrument to apply torque to the anterior actuator nut **24**. The driver recess for the anterior actuator nut **24** may be preferably different from the driver recess **146** for the actuator screw **22**. The distal end **152** of the actuator nut **24** may have an enlarged collar **156**. An anterior drag ring **160**, such as a PEEK washer or annular ring, may be nested against the collar **156**. The anterior drag ring **160** may be captured between the anterior actuator nut **24** and anterior actuator **28** to provide frictional resistance against back-driving the anterior expansion mechanism resulting in loss of anterior height of the implant. The anterior actuator nut **24** may be permanently captured inside the anterior actuator **28**, for example, by two pins **162** or other suitable mechanism.

(41) The posterior actuator nut **26** includes a body with a central cylindrical bore **170** defined therethrough. The bore **170** defines internal threads which are configured to threadably engage with the exterior threads of the distal tip **142** of the actuator screw **22**. The actuator screw **22** is captured inside the stationary posterior actuator ramp **32** by threading into the posterior actuator nut **26**. The end of the posterior actuator nut **26** may have an enlarged collar **172** such that a posterior drag ring **180** may be seated beneath the collar **172**. The posterior drag ring **180**, such as a PEEK washer or annular ring, is captured between the posterior actuator nut **26** and stationary posterior actuator ramp **32** to provide frictional resistance against back-driving the posterior expansion mechanism resulting in loss of overall expanded height of the implant **10**. The drag rings **160**, **180** may be added between the actuator screw **22** and the stationary posterior actuator **32** and between the anterior actuator **28** and anterior actuator nut **24** to ensure that neither the anterior or posterior actuators **28**, **32** lose height during use.

(42) During operation, the implant **10** may be operated in one of two modes. In a first mode, the actuator screw **22** and the anterior actuator nut **24** are rotated or turned together simultaneously. This moves the moveable posterior actuator **30** toward the stationary posterior actuator **32**, forcing the upper and lower endplates **16**, **18** apart. Because the ramp angle of the ramps **72** on the anterior end of the endplate **16**, **18** match the ramp angle of the ramps **70** on the posterior end of the

endplate **16**, **18**, this results in equal expansion of both endplates **16**, **18**. For example, FIGS. **2**A-**2**B show the endplates **16**, **18** expanded in a parallel manner. In a second mode, the actuator screw **22** is held in place and the anterior actuator nut **24** is rotated or turned alone. This makes the anterior actuator **28** move alone which expands or contracts the anterior end of each endplate only and results in an increase in lordotic angle. For example, FIGS. **3**A-**3**B show the endplates **16**, **18** expanded with an increased anterior height.

(43) The expansion mechanism simplifies the expansion driver (not shown) because the actuator screw **22** and the anterior actuator nut **24** need to be turned in the same direction for both lordotic and parallel expansion. In other designs, the components **22**, **24** are rotated in opposite directions, for example, if the actuator screw needed to be turned clockwise for parallel expansion, the anterior actuator nut needed to be turned counterclockwise for lordotic expansion. This required either counterintuitive counter-clockwise motion for implant expansion, or a complex and expensive driver mechanism. Conversely, in this embodiment, the actuator screw **22** and the anterior actuator nut **24** are rotated in the same direction for both lordotic and parallel expansion, thereby simplifying the operation.

(44) The implant **10** allows for continuous expansion and distraction over the range of that specific implant. This provides the ability to distract vertebral bodies to a desired height, but also collapse the device **10** for repositioning if desired. The implant **10** has the ability for the endplates **16**, **18** to converge providing lordosis, while maintaining a large window for bone graft placement. By changing lordotic angulation, the implant **10** may match the patient's natural lordosis or be used to provide a specific lordosis at the level(s) treated.

(45) Turning now to FIGS. **8**A-**13**B, an expandable fusion device or implant **210** is shown according to one embodiment. Implant **210** is similar to implant **10** with the addition of separate anterior and posterior pivot ramps **234**, **236**, **238**. The expandable implant **210** extends along a central longitudinal axis A between front end or posterior end **212** and rear end or anterior end **214** of the device **210**. The implant **210** includes upper and lower endplates **216**, **218** configured to engage adjacent vertebrae, which define a height of the implant **210**. The implant **210** includes an actuator assembly **220** configured for expanding the upper and lower endplates **216**, **218**.

(46) The actuator assembly **220** includes a single actuator screw **222** threaded into anterior and posterior actuator ramps **228**, **230** to independently control the anterior and posterior height of the implant. The implant **210** may include three separate actuators **228**, **230**, **232**: a moveable anterior actuator **228**, a moveable posterior actuator **230**, and a stationary posterior actuator **232**. One end of the actuator screw **222** threads into an anterior actuator nut **224** located in the moveable anterior actuator **228** and the opposite end of the actuator screw **222** threads into a posterior actuator nut **226** affixed to the stationary posterior actuator **232**. The actuator screw **222** threads through the moveable posterior actuator **230** to translate the posterior actuator **230** along longitudinal axis A. The implant **210** also includes two anterior pivot ramps **234** slid onto each side of the anterior actuator **228**, which interface with the upper and lower endplates **216**, **218**. The anterior pivot ramps **234** translate axial motion from the actuator screw **222** into anterior expansion, and pivot on the anterior actuator **228** to allow for independent expansion of the anterior and posterior actuators **228**, **230**, **232**. The posterior expansion mechanism functions by using the actuator screw **222** to drive the moveable posterior actuator **230** toward the stationary posterior actuator **232**. As these two components **230**, **232** move toward one another, ramped surfaces **312**, **332** push the posterior pivot ramps **236**, **238** out in both directions, expanding the posterior height of the implant **210**.

(47) As best seen in FIGS. **8**A-**8**B, the implant **210** is configured to be inserted into the disc space in a collapsed configuration. Once inserted into the disc space, the implant **210** is expanded in height to an expanded configuration to precisely restore spinal alignment and distribute load across the vertebral endplates. FIGS. **9**A-**9**B show the implant **210** in a fully expanded configuration. FIGS. **10**A-**10**B show the implant **210** in an expanded configuration where the anterior and posterior heights are independently adjustable to a desired lordotic profile. In this manner, the

height is adjustable to restore height loss in the disc space and lordotic angulation.

(48) Similar to endplates **16**, **18**, endplates **216**, **218** include front or posterior rail **240** and rear or anterior rail **242** extending between opposed side rails **244**, **246**, which define an inner face **250** and an opposite outer face **252**. The inner face **250** may be configured to mate with the pivot ramps **234**, **236**, **238** and the outer face **252** may be configured to contact adjacent vertebrae, for example, with teeth **254** or other friction increasing elements. Each endplate **216**, **218** may define a vertical window or through passage **256** to define a central graft chamber configured to receive graft material or other therapeutic material.

(49) In the same manner as implant **10**, implant **210** may be secured to the adjacent vertebrae with one or more anchors or fixation screws (not shown) through respective anchor sockets **60**. It will be appreciated that the sockets **60** may be present in any suitable number and configuration for fixation. In the alternative, the sockets **60** may be omitted to provide a standalone device. Cam style blocking screws **62** may be secured in blocking screw holes **64** in the endplates **216**, **218**, which block the anchors or fixation screws from migrating or backing out. In addition, the implant **210** may include one or more recesses or openings **66** for receiving an instrument, such as an insertion and/or expansion instrument.

(50) The endplates **216**, **218** are configured to expand via anterior pivot ramps **234** on the anterior or rear end **214** and posterior pivot ramps **236**, **238** on the posterior or front end **212** of the implant **210**. To accommodate the posterior pivot ramps **236**, **238**, the inner face **250** of each endplate **216**, **218** may include a lateral groove or channel **270** configured to receive the posterior pivot ramps **236**, **238** therein. The channel **270** may be positioned along the posterior rail **240** and defined into the inner face **250** of each endplate **216**, **218**. The channel **270** may be generally perpendicular to the longitudinal axis A of the implant **210**. The channel **270** may have a rounded or curved profile to receive a corresponding protrusion or rib **388** from the posterior pivot ramps **236**, **238**. For example, the channel **270** may define a semicircular, hemispherical, parabolic, or other suitable cross-section.

(51) The anterior end **214** of each endplate **216**, **218** may include one or more ramps **272** configured to mate with actuator **228**, thereby causing the anterior end **214** of the endplates **216**, **218** to separate apart. For example, the endplates **216**, **218** may define anterior ramps **272** along the inner face **250** of the side rails **244**, **246** toward the anterior end **214**. The ramps **272** may include ramped surfaces, angled surfaces, or inclined planes with a given gradient or angle of slope. The ramps **272** may have generally straight ramped surfaces, may be curved, or may be configured in any suitable manner for slidable interface between the components. The ramps **272** may define male or female slide ramps configured to mate with corresponding ramps **294** on the anterior pivot ramps **234**. As best seen in FIG. **13**B, the ramps **272** on endplates **216**, **218** include female ramps with grooves or slots **274** defined therein, such as T-slots. The groove or slots **274** may be configured to receive a portion of anterior pivot ramps **234**, which pivot about actuator **228**. The T-shaped engagement on these anterior pivot ramps **234** interface with the T-slots **274** on the upper and lower endplates **216**, **218** and prevent disassembly. These anterior pivot ramps **234** translate axial motion from the actuator screw **222** into anterior expansion, and their ability to pivot on the anterior actuator **228** allows for independent expansion of the anterior and posterior actuators **228**, **230**, **232**.

(52) The actuator assembly **220** includes moveable anterior actuator **228** with anterior pivot ramps **234** positioned between the upper and lower endplates **216**, **218**, thereby providing anterior expansion to the implant **210**. The moveable anterior actuator **228** includes a laterally extending body with an enlarged central portion **280**. The anterior actuator **228** extends from a first free end **282** to a second free end **284**. The first and second ends **282**, **284** may each define a cylinder with a circular cross-sectional shape configured to receive the anterior pivot ramps **234**. The first and second free ends **282**, **284** are receivable between the side rails **244**, **246** toward the anterior end **214** of the upper and lower endplates **216**, **218** and the enlarged central portion **280** is positionable

in the graft window **256** of the upper and lower endplates **216**, **218** when in the collapsed configuration. The enlarged central portion **280** defines a central non-threaded bore **286** configured to receive the anterior actuator nut **224**. The central axis of bore **286** may be aligned with the central longitudinal axis A of the implant **210**. Similar to anterior actuator **28**, additional recesses or bores **88** may be defined into the anterior actuator **228**. For example, a first threaded bore **88** may be provided on one side of the central bore **286** and a second non-threaded bore **88** may be provided on the opposite side of the central bore **286**, which are configured to attach an instrument, such as an insertion and/or expansion instrument to the anterior actuator **228**. It will be appreciated that any suitable number, type, and configuration of instrument recesses may be used.

(53) The moveable anterior actuator **228** supports one or more anterior pivot ramps **234** configured to mate with corresponding ramps **272** on the endplates **216**, **218**. Each anterior pivot ramp **234** may include a ring **288** and a foot **290**. The ring **288** may be a full annular ring defining a central cylindrical through hole **292**. The axis of through hole **292** may be aligned with a central lateral axis between ends **282**, **284** of the anterior actuator **228**. The ring **288** may have an outer surface and an inner surface configured to contact the anterior actuator **228**. In particular, the cylindrical ends **282**, **284** of anterior actuator **228** are positionable through the cylindrical through holes **292** of each anterior pivot ramps **234**, thereby providing for pivotal movement about the anterior actuator **228**. For example, two anterior pivot ramps **234** may be slid onto each cylindrical end **282**, **284** of anterior actuator **228**. The smooth inner surface of the anterior pivot ramps **234** allows each pivot ramp **234** to freely rotate on the corresponding smooth outer cylindrical surface of the anterior actuator **228**.

(54) The anterior pivot ramps **234** include foot **290** with a sliding surface **294** configured to mate with a corresponding sliding surface within groove **274** in the endplate **216**, **218**. The foot **290** may be a male projection extending away from one side of ring **288**. The foot **290** may have a channel or groove **296** around its periphery, thereby forming the male projection, which is configured to enter the corresponding female recess **274** in the endplate **216**, **218**. The foot **290** may define an arched surface on one end and a planar surface on the opposite end. The sliding surface **294** may be a smooth flat or planar surface. The groove **296** may cause portions of the foot **290**, including the arched end, to form an overhang. For example, the foot **290** may have a generally T-shaped cross section. The T-shaped foot **290** on the anterior pivot ramps **234** interface with the T-slots **274** on the upper and lower endplates **216**, **218** and prevent disassembly.

(55) The sliding surface **294** of each anterior pivot ramp **234** mates with the corresponding sliding surfaces within the grooves **274** in the endplates **216**, **218** to allow for sliding motion between the components. For example, two pivot ramps **234** may be positioned on one end **282** of the anterior actuator **228**: one configured to slidably engage the upper endplate **216** and one configured to slidably engage the lower endplate **218**. Similarly, two pivot ramps **234** may be positioned on the opposite end **284** of the anterior actuator **228**: one configured to slidably engage the upper endplate **216** and one configured to slidably engage the lower endplate **218**. The anterior pivot ramps **234** are able to freely rotate on the cylindrical surfaces of the anterior actuator **228**, which provides continuous contact between the sliding surfaces **294** of the anterior pivot ramps **234** and the endplates **216**, **218** when the angulation between the endplates **216**, **218** changes. The anterior pivot ramps **234** translate axial motion from the actuator screw **222** into anterior expansion, and their ability to pivot on the anterior actuator **228** allows for independent expansion of the anterior and posterior heights.

(56) The actuator assembly **220** includes moveable posterior actuator **230** positioned between the upper and lower endplates **216**, **218**, thereby providing posterior expansion to the implant **210**. The moveable posterior actuator **230** includes a laterally extending body connecting a first free end **302** to a second free end **304** with a flat anterior face **306** and an opposite posterior face **308**. The first and second free ends **302**, **304** are receivable between the upper and lower endplates **216**, **218** toward the posterior end **212** of the implant **210**. The posterior actuator **230** defines a central

cylindrical threaded bore **310** configured to receive the actuator screw **222** therethrough. The central axis of bore **310** may be aligned with the central longitudinal axis A of the implant **210**. A raised collar may be provided around bore **310** to reinforce the region surrounding the bore **310**, thereby facilitating threaded engagement with the actuator screw **222** or additional strength to the posterior actuator **230**.

(57) The moveable posterior actuator **230** includes one or more ramps **312** configured to mate with corresponding ramps **394** on the posterior pivot ramps **236**, **238**. The ramps **312** may be defined into and/or extend from the posterior face **308** of the posterior actuator **230**. The ramps **312** may include ramped surfaces, angled surfaces, or inclined planes with a given gradient or angle of slope. The ramps **312** may have generally straight ramped surfaces, may be curved, or may be configured in any suitable manner for slidable interface between the components. The ramps **312** may define male or female slide ramps configured to mate with corresponding ramps **394** on the posterior pivot ramps **236**, **238**. Similar to foot **290** on the anterior pivot ramps **234**, the ramps **312** may include a sliding surface configured to mate with a corresponding sliding surface within groove **394** in the posterior pivot ramps **236**, **238**. The ramps **312** may be a male projection extending posteriorly. The ramps **312** may have a channel or groove around its periphery, thereby forming the male projection, which is configured to enter the corresponding female recess **394** in the posterior pivot ramps **236**, **238**. The ramps **312** may have a generally T-shaped cross section. The T-shaped ramps **312** on the moveable posterior actuator **230** interface with the T-slots **394** on the posterior pivot ramps **236**, **238** and prevent disassembly and allow for independent expansion. Although a T-shaped interface is exemplified, it will be appreciated that other suitable slidable mating interfaces may also be used.

(58) The actuator assembly **220** includes stationary posterior actuator **232**, which defines the nose or front end **212** of the implant **210** and provides posterior expansion to the implant **210** using the posterior pivot ramps **236**, **238**. The stationary posterior actuator **232** includes a laterally extending body having a first free end **320** and a second free end **322** on opposite sides. The stationary posterior actuator **232** includes an anterior side **324** and an opposite posterior side **326**. The anterior side **324** may be split into two angled faces: an upper portion tilted upward, and a lower portion tilted downward. The anterior side **324** defines a central non-threaded bore **330** configured to receive the posterior actuator nut **226**. The central axis of bore **330** may be aligned with the central longitudinal axis A of the implant **210**. A lateral spine with a central raised collar may be provided around bore **330** to reinforce the region surrounding the bore **330**, thereby facilitating engagement with posterior actuator nut **226** or additional strength to the posterior actuator **232**.

(59) The stationary posterior actuator **232** includes one or more ramps **332** configured to mate with corresponding ramps **394** on posterior pivots ramps **236**, **238**. The ramps **332** may be defined into and/or extend from the anterior face **324** of the stationary posterior actuator **232**. The ramps **332** may include ramped surfaces, angled surfaces, or inclined planes with a given gradient or angle of slope. The ramps **332** may have generally straight ramped surfaces, may be curved, or may be configured in any suitable manner for slidable interface between the components. The ramps **332** may define male or female slide ramps configured to mate with corresponding ramps **394** on the posterior pivot ramps **236**, **238**. Similar to ramps **312**, the ramps **332** may include a sliding surface configured to mate with a corresponding sliding surface within groove **394** in the posterior pivot ramps **236**, **238**. The ramps **332** may be male projections extending anteriorly, with two ramps **332** extending upward from the upper angled face and two ramps **332** extending downward from the lower angled face. The ramps **332** may have a channel or groove around its periphery, thereby forming the male projection, which is configured to enter the corresponding female recess **394** in the posterior pivot ramps **236**, **238**. The ramps **332** may have a generally T-shaped cross section or other suitable shape, which interface with the T-slots **394** on the posterior pivot ramps **236**, **238**.

(60) The stationary posterior actuator **232** may include one or more cutouts **334** configured to receive portions of the upper and lower posterior pivot ramps **236**, **238** when the implant **10** is in its collapsed position. As shown in FIG. **8**B, in the collapsed position, the anterior face **306** of the

moveable posterior actuator **230** may contact or abut the posterior face of the anterior actuator **228**. The posterior expansion mechanism functions by using the actuator screw **222** to drive the moveable posterior actuator **230** toward the stationary posterior actuator **232**. FIG. **9**B shows the moveable posterior actuator **230** separated from anterior actuator **228** and moved posteriorly toward stationary posterior actuator **232**. As the two posterior actuators **230**, **232** move toward one another, these actuators **230**, **232** interface with ramped surfaces **394** on the posterior pivot ramps **236**, **238**, pushing the upper and lower endplates **216**, **218** out in both directions, expanding the posterior height of the implant **210**.

(61) The upper and lower posterior pivot ramps **236**, **238** may be the same or similar with mirrored and/or reversed positions. Each of the posterior pivot ramps **236**, **238** may be separated into left and right sections **380**, **382** positioned on opposite sides of the central actuator screw **222**. Each section **380**, **382** may include a laterally extending body configured to mate with the respective posterior actuators **230**, **232** and the upper and lower endplates **216**, **218**. As best seen in FIG. **9**B, the posterior pivot ramps **236**, **238** may have a triangular or bell-shaped cross section with a narrower or tapered end **384** facing inward and a wider or flared end **386** facing outward. The flared end **386** has a lateral protrusion, lip, or rib **388** extending generally perpendicular to the longitudinal axis A of the implant **210**. Each lateral rib **388** is receivable in the respective groove or channel **270** in the upper and lower endplates **216**, **218**. The lateral rib **388** may have a rounded or partially circular cross section configured to pivot in the respective channel **270**. The circular rib **388** on the posterior pivot ramps **236**, **238** interfaces with the corresponding cut **270** in the upper and lower endplates **216**, **218** preventing disassembly and allowing for independent expansion of the anterior and posterior heights.

(62) The upper and lower posterior pivot ramps **236**, **238** have an anterior-facing side **390** and a posterior-facing side **392**. The anterior-facing and posterior-facing sides **390**, **392** may be slanted or sloped between the tapered end **384** and the flared end **386**. The anterior-facing and posterior-facing sides **390**, **392** define one or more ramps **394** configured to mate with the corresponding ramps **312**, **332** on the posterior actuators **230**, **232**. The ramps **394** may be defined into and/or extend from the anterior-facing and posterior-facing sides **390**, **392** of the posterior pivot ramps **236**, **238**. The ramps **394** may include ramped surfaces, angled surfaces, or inclined planes with a given gradient or angle of slope. The ramps **394** may have generally straight ramped surfaces, may be curved, or may be configured in any suitable manner for slidable interface between the components. Similar to ramps **272**, the ramps **394** on posterior pivot ramps **236**, **238** include female ramps with grooves or slots defined therein, such as T-slots. The T-shaped engagement on these posterior actuators **230**, **232** interface with the T-slots **384** on the posterior pivot ramps **236**, **238**. As the moveable posterior actuator **230** moves toward the stationary posterior actuator **232**, the ramped surfaces **312**, **332**, **394** push the left and right posterior pivot ramps **380**, **382** out in both directions, expanding the posterior height of the implant **210**.

(63) The actuator assembly **220** includes actuator screw **222**, anterior actuator nut **224**, and posterior actuator nut **226** aligned along the central longitudinal axis A of the implant **210**. The actuator screw **222** extends from a proximal end **340** to a distal end **342**. The actuator screw **222** may include a shaft with an exterior threaded portion **344** extending along its length. The threaded shaft **344** may have a given diameter, handedness, thread form, thread angle, lead, pitch, etc. suitable for interfacing with both the anterior actuator nut **224** and the moveable posterior actuator **230**. In this embodiment, a single type of thread profile is used for both parallel and lordotic expansion. Other systems may require three different sets of threads to engage for parallel expansion. Conversely, threaded shaft **344** achieves parallel and lordotic expansion through the interaction of one set of threads. This reduces the overall increase in lifting torque created by friction or binding of the threads.

(64) The proximal end **340** may define an instrument recess **346** configured to receive an instrument, such as a driver, to rotate or actuate the actuator screw **222**. The instrument recess **346**

may include a tri-lobe, hex, star, or other suitable recess configured to engage with a driver instrument to apply torque to the actuator screw **222**. The proximal end **340** of the shaft is configured to thread into the anterior actuator nut **224** to translate the moveable anterior actuator **228**. The threaded shaft **344** also threads through the moveable posterior actuator **230** to translate the moveable posterior actuator **230**.

(65) The distal end **342** of the actuator screw **222** may have a reduced distal tip, for example, having a diameter less than the diameter of the threaded shaft **344**. The distal tip **342** may be threaded and configured to thread into the posterior actuator nut **226**. Even at full expansion, for example shown in FIGS. **9**A-**9**B, the distal end **342** of the actuator screw **222** does not protrude out past the posterior edge or front nose **212** of the implant **210** as may be seen in other designs. Posterior protrusion of the actuator screw was a point of concern from surgeons due to the optics of having the rod protruding near the posterior structures of the spine. Thus, the expansion assembly **220** for implant **210** eliminates the possibility of any posterior protrusion of the actuator screw **222**.

(66) The anterior actuator nut **224** has a body extending between a proximal end **350** to a distal end **352**. As best seen in FIG. **13**A, a central bore **354** extends through the body of the anterior actuator nut **224** from the proximal end **350** to the distal end **352**. A portion of the bore **354** defines internal threads which are configured to threadably engage with the exterior threads **344** of the actuator screw **222**. The proximal end **350** of the anterior actuator nut **224** defines a driver engagement recess, such as a series of notches and teeth or other suitable recess configured to engage with a driver instrument to apply torque to the anterior actuator nut **224**.

(67) The distal end **352** of the actuator nut **224** may have an enlarged collar **356**. Similar to implant **10**, an anterior drag ring **160**, such as a PEEK washer or annular ring, may be nested against the underside of the collar **356**. The anterior drag ring **160** may be captured between the anterior actuator nut **224** and anterior actuator **228** to provide frictional resistance against back-driving the anterior expansion mechanism, which could result in loss of anterior height of the implant. The anterior actuator nut **224** may be permanently captured inside the anterior actuator **228**, for example, by two pins **162** or other suitable mechanism. Additionally, a mechanical lock **360** may sit in the anterior actuator **228** and interface with the teeth of the anterior actuator nut **224** to provide additional security against loss of height of the implant **210**.

(68) The posterior actuator nut **226** includes a body with a central cylindrical bore **370** defined therethrough. The bore **370** defines internal threads which are configured to threadably engage with the exterior threads of the distal tip **342** of the actuator screw **222**. The actuator screw **222** is captured inside the stationary posterior actuator ramp **232** by threading into the posterior actuator nut **226**. The end of the posterior actuator nut **226** may have an enlarged collar **372** such that posterior drag ring **180** may be seated beneath the collar **372**. Similar to implant **10**, the posterior drag ring **180**, such as a PEEK washer or annular ring, is captured between the posterior actuator nut **226** and stationary posterior actuator ramp **232**. The drag ring **180** may provide frictional resistance against back-driving the posterior expansion mechanism resulting in loss of overall expanded height of the implant **210**.

(69) During operation, the implant **210** may be operated in one of two modes. In a first mode, the actuator screw **222** and the anterior actuator nut **224** are rotated and turned together at the same time. This moves the moveable posterior actuator **230** posteriorly toward the stationary posterior actuator **232**, forcing the posterior pivot ramps **236**, **238** upward. This results in equal, parallel expansion of both endplates **216**, **218**. In a second mode, the actuator screw **222** is not turned and the anterior actuator nut **224** is rotated or turned by itself. This makes the anterior actuator **228** move alone, which expands the anterior end of each endplate **216**, **218** only and results in an increase in lordotic angle.

(70) Similar to implant **10**, implant **210** allows continuous expansion and distraction over the range of that specific implant. The vertebral bodies may be distracted to a desired height and the implant may be collapsed for repositioning, if desired. During expansion, the implant endplates may be

expanded in parallel or may converge to provide lordosis, while maintaining a large window for bone graft placement. By changing lordotic angulation, the implant may be expanded to match the patient's natural lordosis or used to provide a specific lordosis at the level(s) treated.
(71) Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that all components of the various devices disclosed above may be combined or modified in any suitable configuration.

Claims

1. An expandable implant comprising: an upper endplate and a lower endplate configured to engage adjacent vertebrae; and an actuator assembly extending along a longitudinal axis of the implant and positioned between the upper and lower endplates for expanding the upper and lower endplates to independently control anterior and posterior heights of the implant, the actuator assembly including a moveable anterior actuator, a moveable posterior actuator, a stationary posterior actuator, an actuator screw threaded into an anterior actuator nut located in the moveable anterior actuator, and a posterior actuator nut located in the stationary posterior actuator, wherein the actuator screw threads through the moveable posterior actuator to translate the moveable posterior actuator, wherein the upper and lower endplates each define a plurality of ramps configured to mate with corresponding male protrusions on the moveable anterior actuator and the moveable and stationary posterior actuators, wherein the male protrusions on the moveable anterior actuator and the moveable and stationary posterior actuators include point contact pivoting ramps defined thereon which act as a pivot point, facilitating pivoting or rotational movement for a lordotic angle adjustment between the upper and lower endplates, wherein when the actuator screw and the anterior actuator nut are turned together, the moveable posterior actuator moves toward the stationary posterior actuator, forcing the upper and lower endplates apart resulting in parallel expansion, and when only the anterior actuator nut is turned, the moveable anterior actuator moves alone to increase the anterior height, resulting in an increase in a lordotic angle of the implant.
2. The expandable implant of claim 1, wherein the ramps on the upper and lower endplates include female T-slots.
3. The expandable implant of claim 1, wherein the moveable anterior actuator includes a laterally extending body with an enlarged central portion defining a non-threaded bore and opposing free ends defining an irregular cross-sectional shape.
4. The expandable implant of claim 1, wherein the moveable posterior actuator includes a laterally extending body defining a threaded bore with a flat anterior face and an opposite posterior face.
5. The expandable implant of claim 1, wherein the stationary posterior actuator includes a laterally extending body defining a non-threaded bore with an anterior face.
6. The expandable implant of claim 1, wherein the actuator screw includes a shaft having a single type of thread profile along its length and a reduced diameter at its distal end for threading into the posterior actuator nut, wherein the single type of thread profile is used to achieves both parallel and lordotic expansion.
7. An expandable implant comprising: an upper endplate and a lower endplate configured to engage adjacent vertebrae; and an actuator assembly extending along a longitudinal axis of the implant and positioned between the upper and lower endplates for expanding the upper and lower endplates to independently control anterior and posterior heights of the implant, the actuator assembly including a moveable anterior actuator receiving and securing a plurality of anterior pivot ramps, a moveable posterior actuator, a stationary posterior actuator engaged with upper and lower posterior pivot

ramps, and an actuator screw threaded into an anterior actuator nut located in the moveable anterior actuator, and a posterior actuator nut located in the stationary posterior actuator, wherein the actuator screw threads through the moveable posterior actuator to translate the moveable posterior actuator, wherein each anterior pivot ramp has a smooth inner surface to allow each anterior pivot ramp to rotate on the moveable anterior actuator, wherein the upper and lower endplates each define a plurality of ramps configured to mate with corresponding ramped surfaces on the anterior pivot ramps and the upper and lower posterior pivot ramps, wherein when the actuator screw and the anterior actuator nut are turned together, the moveable posterior actuator moves toward the stationary posterior actuator, forcing the upper and lower posterior pivot ramps outward resulting in parallel expansion of the upper and lower endplates, and when only the anterior actuator nut is turned, the moveable anterior actuator moves alone to increase the anterior height, resulting in an increase in a lordotic angle of the implant.
8. The expandable implant of claim 7, wherein each anterior pivot ramp has a ring received on the moveable anterior actuator and a foot comprising the ramped surface configured to mate with the respective ramps on the upper and lower endplates.
9. The expandable implant of claim 8, wherein each foot is a male projection extending from one side of the ring, and each ramp on the upper and lower endplates is female recess configured to receive the respective male projection of the foot.
10. The expandable implant of claim 8, wherein the moveable anterior actuator includes a laterally extending body with an enlarged central portion defining a non-threaded bore and opposing free ends defining cylindrical ends.
11. The expandable implant of claim 10, wherein each anterior pivot ramp is configured to rotate on the cylindrical ends of the moveable anterior actuator.
12. The expandable implant of claim 7, wherein the upper and lower posterior pivot ramps each include left and right sections positioned on opposite sides of the actuator screw.
13. The expandable implant of claim 7, wherein the upper and lower posterior pivot ramps each include a laterally extending body with a bell-shaped cross section having a tapered end and a flared end.
14. The expandable implant of claim 13, wherein the flared end of the upper and lower posterior pivot ramps includes a lateral rib having a circular cross section configured to pivot in a corresponding channel in the upper and lower endplates.
15. A method of spinal fixation comprising: inserting an expandable intervertebral implant into a disc space between adjacent vertebrae, the expandable implant comprising upper and lower endplates each configured to engage one of the adjacent vertebrae and an actuator assembly extending along a longitudinal axis of the implant and positioned between the upper and lower endplates for expanding the upper and lower endplates to independently control anterior and posterior heights of the implant, the actuator assembly including a moveable anterior actuator, a moveable posterior actuator, a stationary posterior actuator, an actuator screw threaded into an anterior actuator nut located in the moveable anterior actuator, and a posterior actuator nut located in the stationary posterior actuator, wherein the actuator screw threads through the moveable posterior actuator to translate the moveable posterior actuator, wherein the upper and lower endplates each define a plurality of ramps configured to mate with corresponding male protrusions on the moveable anterior actuator and the moveable and stationary posterior actuators, wherein the male protrusions on the moveable anterior actuator and the moveable and stationary posterior actuators include point contact pivoting ramps defined thereon which act as a pivot point, facilitating pivoting or rotational movement for a lordotic angle adjustment between the upper and lower endplates; and expanding the implant in height by: (1) rotating the actuator screw and the anterior actuator nut together to move the moveable posterior actuator toward the stationary posterior actuator, thereby forcing the upper and lower endplates apart resulting in parallel expansion; or (2) rotating only the anterior actuator nut to move the moveable anterior actuator

alone to increase the anterior height, resulting in an increase in a lordotic angle of the implant.

16. The method of claim 15, further comprising inserting bone anchors through sockets in the upper and lower endplates and into the adjacent vertebrae.

17. The method of claim 16, further comprising inserting blocking screws through holes in the upper and lower endplates and rotating the blocking screws in the upper and lower endplates to cover the bone anchors and prevent the bone anchors from backing out.